

1-DoF Helicopter

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Abstract—

Index Terms—DoF, Helicopter, Non-linear Control, Nonlinear System, Modelling, Sliding Mode Control, Back-Stepping,

I. INTRODUCTION

Unmanned Aerial Vehicles (UAVs), have attracted the attention of scientific research, in both private institutions and hobby researchers, due to their flexible maneuvering capabilities [1], 1 DoF is specifically interesting for research due to its simplified model and ease of study, as it offers a more simplified model of a helicopter, to study the dynamics and control of rotary-wing aircrafts, restricting the motion to a single axis, which is usually demonstrated as pitch and yaw angles, making it appealing for educational purposes as it neglects the complexities of full-scale helicopter dynamics. In literature, different control techniques have been implemented and tested into the system, to control BLDC motor based propulsion systems, such as Sliding Mode Control (SMC), which is widely used in nonlinear systems [2], where it showed more robustness to disturbances when compared to control methods such as the Proportional Integral Derivative (PID) control.

The system's hardware consists of a rigid body attached to a pivot point, allowing rotation in one axis, with a brushless DC motor acting as a propeller, with sensors like the IMU and magnetometer providing feedback on the system's current state, to aid in the stabilizing process.

II. SYSTEM MODELING

In this work, the modeling of a 1 Degree of Freedom (DoF) helicopter is developed to simulate its dynamic behavior for control design. The system consists of a single-rotor helicopter model, represented by a wooden bar connected to a wall via a hinge that allows one degree of freedom in rotation. A brushless DC (BLDC) motor with a propeller is mounted at the far end of the wooden bar, away from the hinge. As the motor rotates, it generates a lifting force that causes the bar to incline from 0 to 90 degrees, simulating the pitch movement

of a helicopter with a fixed horizontal axis, as shown in Fig. 1.

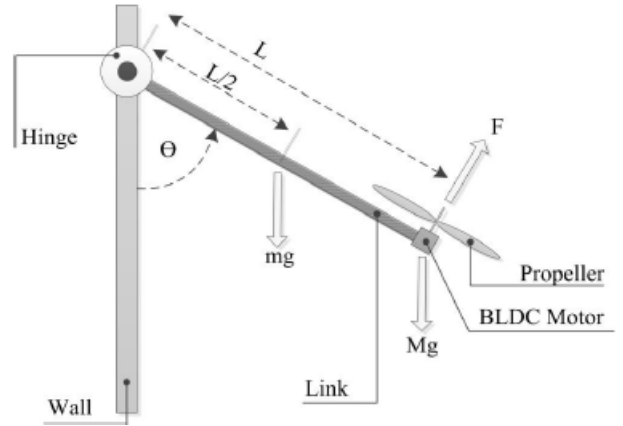


Fig. 1. 1 DoF helicopter system

The forces acting on the system include the lift generated by the propeller and the gravitational forces resulting from the weight of the motor at the end of the bar and the weight of the bar itself, which acts at its center of mass. These forces generate a resultant torque about the hinge, leading to the rotational motion of the system. A mathematical model of the system is derived using Newton's laws of motion, calculating the torque about the hinge. The overall model was divided into 2 separate models, which was then cascaded to give us the overall system model.

A. System Model

The system includes 2 states, the angle of the wooden bar θ and its angular velocity $\dot{\theta}$. The system is governed by the equation:

$$\ddot{\theta} = \frac{FL}{J} - \frac{mgL}{J}\sin(\theta) \quad (1)$$

where F thrust force, which is the system's input force u , L is the length of the lever arm, J is the lever's moment of inertia,

m the motor's mass added to half the lever's mass, g is the gravitational acceleration.

The system's input force F links the thrust with the the w of the motor, using the following relation [3]:

$$F = \frac{1}{2} C_T \pi \rho r^2 v^2 \quad (2)$$

where C_T is the thrust coefficient, ρ is the air's density, r is the radius of the blade's propeller, v is the speed of air.

The velocity v can be used to directly relate the v and ω as follows:

$$v = k\omega \quad (3)$$

and the constant k can be calculated as:

$$k = \frac{\text{pitch}}{2\pi} \quad (4)$$

where pitch is considered to be the distance the propeller drives through air in one revolution.

B. Motor Model

The used brushless DC motor was modeled as a DC motor, for its simplified control design, as the focus is lies in high level control, this simplification can be done by the BLDC's torque characteristics using a similar fashion to the DC motor's speed characteristics. Also, the motor is assumed to operate in its linear region where the back-EMF is approximately linear.

The state space representation of the motor can be modeled as follows:

$$\dot{x} = \begin{bmatrix} -\frac{R}{J} & -\frac{K_e}{J} \\ \frac{K_T}{J} & -\frac{B}{J} \end{bmatrix} \begin{bmatrix} I \\ \omega \end{bmatrix} + \begin{bmatrix} \frac{1}{J} \\ -\frac{1}{J} \end{bmatrix} \begin{bmatrix} V \\ T_L \end{bmatrix} \quad (5)$$

where B is the viscous damping of the motor, K_e is the back emf constant, i is the current drawn by the motor, V is the voltage, and T_L is the torque load.

C. Simulink Model

The system modeling was performed on Simulink, with the parameters being identified in MATLAB scripts, figure 2 shows the modeling blocks.

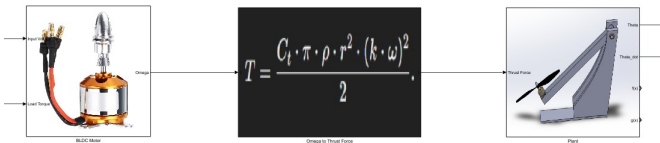


Fig. 2. Simulink system modeling blocks

Figure 3 shows the implemented model on simulink, simulating the system equation explained above and figure 4 shows the the transition from the angular speed w and the thrust force F , where all the parameters are being added to a multiplier.

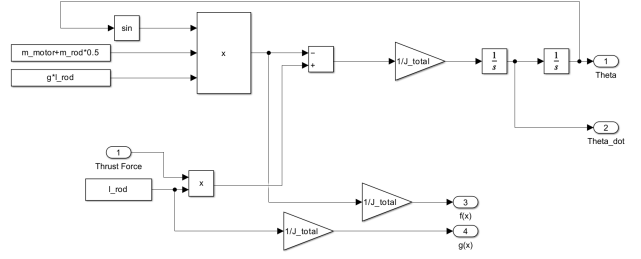


Fig. 3. Inside the plant block

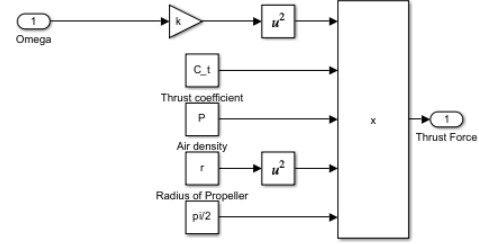


Fig. 4. The thrust block

III. METHODOLOGY

A. System Identification

In order to estimate the unknown parameters J_m , $\tilde{k}$ and the damping constant B , it was required from us to perform system identification to simulate the system's behavior based gathered experimental data by varying the input voltage manually, to get different roll angle readings from the IMU, as well as numerical estimations using 4th order Runge Kutta method, and known parameters from our model.

5 shows the pole placement, 6 shows the system response to an input voltage and the difference between simulated and actual beam angle, which is measured from the IMU sensor, and 7 shows the difference between the simulated and measured beam angular position, and the system's response to input voltage, which shows a close correlation between the input voltage and the angular position θ .

From the code, we estimated the state space matrices with:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -43.2794 & 0 & 0 & 0 \\ 0 & 0 & -0.2980 & 0.297 \\ 0 & 0 & -86.8000 & -86.8000 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 20000 \end{bmatrix}, C = [1 \ 0 \ 0 \ 0], D = [0]$$

and according to the output of the optimization problem, $\tilde{k} = 4.8844$, $J_m = 3.3623$ & $B = 1.0019$.

The system from 7 shows that there is no response to input signal, where the the system should show excitation or response when an input is given to the system.

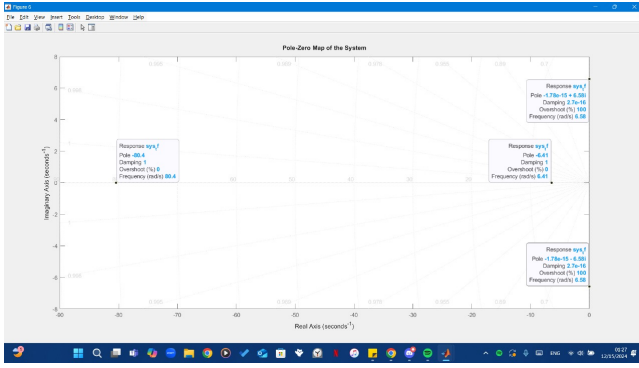


Fig. 5. The pole placement

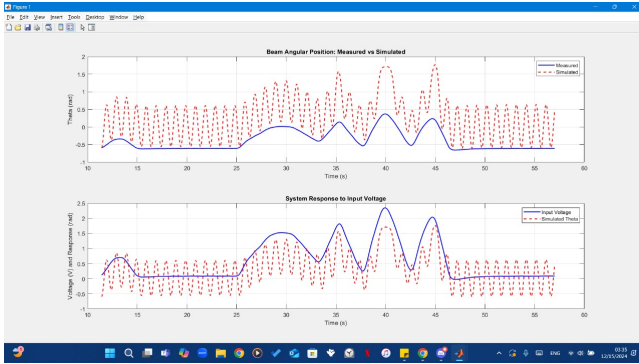


Fig. 6. Difference between measured and estimated values

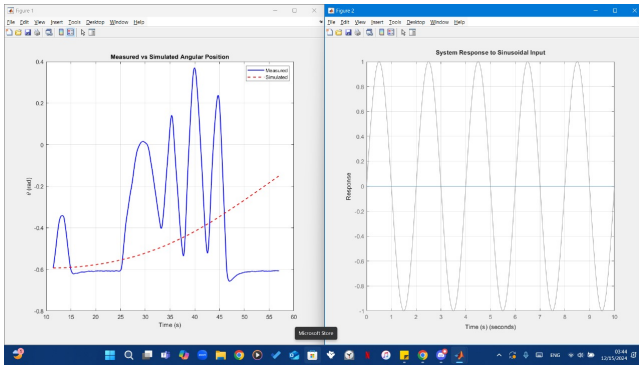


Fig. 7. System response to input

B. Input Output Linearization

1) *Nonlinear System Dynamics:* The system dynamics are described by the following equations:

$$\ddot{\theta} = -\frac{mgl}{J} \sin(\theta) + \frac{l}{J} F, \quad (6)$$

$$\ddot{\theta} = -\frac{mgl}{J} \cos(\theta) \dot{\theta} + \frac{l}{J} \dot{F}, \quad (7)$$

where:

- θ is the angular position of the system,
- $F = K\omega^2$ is the thrust force generated by the motor,
- K is the thrust constant,
- ω is the motor rotational velocity,

- m , g , l , and J are the system parameters representing mass, gravity, length, and moment of inertia, respectively.

The thrust force F is related to the rotational velocity of the motor as:

$$\dot{F} = 2K\omega\dot{\omega}, \quad (8)$$

and higher-order derivatives propagate through the system dynamics.

2) *Control Design Using Virtual Input:* The control strategy involves introducing a virtual input $\ddot{\omega}$ to cancel nonlinear terms and achieve precise trajectory tracking. By differentiating (6) and (7), the fourth derivative of θ is derived:

$$\ddot{\ddot{\theta}} = -\frac{mgl}{J} \cos(\theta) \ddot{\theta} + \frac{l}{J} \ddot{F}, \quad (9)$$

where:

$$\ddot{F} = 2K(\dot{\omega}^2 + \omega\ddot{\omega}). \quad (10)$$

The nonlinear terms $\sin(\theta)$ and $\cos(\theta)$ are explicitly canceled using $\ddot{\omega}$, allowing the system to behave in a linearized manner.

3) *Feedback Control Law:* To achieve tracking of the desired trajectory θ_d , we define the tracking errors:

$$e_\theta = \theta - \theta_d, \quad (11)$$

$$e_{\dot{\theta}} = \dot{\theta} - \dot{\theta}_d, \quad (12)$$

$$e_{\ddot{\theta}} = \ddot{\theta} - \ddot{\theta}_d, \quad (13)$$

$$e_{\ddot{\ddot{\theta}}} = \ddot{\ddot{\theta}} - \ddot{\ddot{\theta}}_d. \quad (14)$$

The control law enforces:

$$\ddot{\ddot{\theta}} = \ddot{\ddot{\theta}}_d - k_1 e_\theta - k_2 e_{\dot{\theta}} - k_3 e_{\ddot{\theta}} - k_4 e_{\ddot{\ddot{\theta}}}, \quad (15)$$

where k_1 , k_2 , k_3 , and k_4 are feedback gains.

4) *Voltage Derivation:* The virtual input $\ddot{\omega}$ is mapped to the control input U (voltage) through the motor dynamics:

$$\ddot{\omega} = \frac{1}{J} (K_t \dot{I} - K_{\tilde{k}} \omega \dot{\omega}), \quad (16)$$

where K_t is the torque constant and $K_{\tilde{k}}$ accounts for nonlinear drag.

The voltage U is computed using the electrical dynamics:

$$U = L\dot{I} + IR + K_e \omega, \quad (17)$$

where L , R , and K_e represent the motor inductance, resistance, and back-EMF constant, respectively.

5) *Advantages of This Approach:*

- **Nonlinear Cancellation:** Nonlinear terms such as $\sin(\theta)$ and $\cos(\theta)$ are explicitly canceled via $\ddot{\omega}$.
- **High-Order Feedback:** The fourth derivative of θ ensures precise trajectory tracking.
- **Exact Dynamics:** No linearization approximations are used, making the controller robust to large deviations.

C. Back-Stepping

Back-stepping control ensures the global asymptotic stability of the system by progressively stabilizing each subsystem: beam angle, angular velocity, and rotor speed. The choice of Lyapunov functions and control inputs guarantees that all error variables (z_1, z_2, z_3) converge to zero.

The Back-stepping control algorithm is applied to stabilize the 1-DOF copter system, ensuring that the beam remains horizontal. The control design proceeds in a step-by-step manner as follows:

The system dynamics are described as:

$$\dot{\theta} = \omega, \quad (18)$$

$$\ddot{\theta} = \frac{-mgL \sin(\theta) + FL}{J}, \quad (19)$$

where $F = K\omega^2$ is the thrust force generated by the rotor. The motor dynamics are given by:

$$L \frac{dI}{dt} = -RI - K_e\omega + V, \quad (20)$$

$$J_m \frac{d\omega}{dt} = K_t I - B\omega - T_L. \quad (21)$$

- Step 1: Stabilize Beam Angle Define the error variable for the beam angle:

$$z_1 = \theta - \theta_d. \quad (22)$$

The Lyapunov function is chosen as:

$$V_1 = \frac{1}{2}z_1^2. \quad (23)$$

Taking the derivative:

$$\dot{V}_1 = z_1 \dot{z}_1 = z_1(\dot{\theta} - \dot{\theta}_d). \quad (24)$$

Substituting $\dot{\theta} = \omega$, the virtual control input ν_1 is designed as:

$$\nu_1 = \dot{\theta}_d - k_1 z_1, \quad (25)$$

where $k_1 > 0$ ensures stability.

- Step 2: Stabilize Angular Velocity Define the error variable for angular velocity:

$$z_2 = \omega - \nu_1. \quad (26)$$

Augment the Lyapunov function:

$$V_2 = V_1 + \frac{1}{2}z_2^2. \quad (27)$$

Taking the derivative:

$$\dot{V}_2 = \dot{V}_1 + z_2 \dot{z}_2 \quad (28)$$

$$= -k_1 z_1^2 + z_2 \left(\frac{-mgL \sin(\theta) + FL}{J} - \dot{\nu}_1 \right). \quad (29)$$

The control input for F is designed as:

$$F = \frac{J}{L}(\dot{\nu}_1 - k_2 z_2 + mg \sin(\theta)), \quad (30)$$

where $k_2 > 0$ ensures stability.

- Stabilize Rotor Speed Define the error variable for rotor speed:

$$z_3 = \omega - \sqrt{\frac{F}{K}}. \quad (31)$$

Augment the Lyapunov function:

$$V_3 = V_2 + \frac{1}{2}z_3^2. \quad (32)$$

Taking the derivative:

$$\dot{V}_3 = \dot{V}_2 + z_3 \dot{z}_3 \quad (33)$$

$$= -k_1 z_1^2 - k_2 z_2^2 + z_3 \left(\frac{K_t I - B\omega}{J_m} - \frac{\dot{F}}{2K} \right). \quad (34)$$

The control input for current I is designed as:

$$I = \frac{J_m}{K_t} \left(\frac{\dot{F}}{2K} + k_3 z_3 + B\omega \right), \quad (35)$$

where $k_3 > 0$ ensures stability.

D. Pole Placement for Gain Selection

To place the closed-loop poles at the desired locations:

$$s_1 = -2, \quad s_2 = -3, \quad s_3 = -4, \quad s_4 = -5,$$

the characteristic polynomial is:

$$(s - s_1)(s - s_2)(s - s_3)(s - s_4) = s^4 + a_1 s^3 + a_2 s^2 + a_3 s + a_4,$$

where:

$$a_1 = -(s_1 + s_2 + s_3 + s_4), \quad (36)$$

$$a_2 = s_1 s_2 + s_1 s_3 + s_1 s_4 + s_2 s_3 + s_2 s_4 + s_3 s_4, \quad (37)$$

$$a_3 = -(s_1 s_2 s_3 + s_1 s_2 s_4 + s_1 s_3 s_4 + s_2 s_3 s_4), \quad (38)$$

$$a_4 = s_1 s_2 s_3 s_4. \quad (39)$$

For the chosen poles:

$$s_1 = -2, \quad s_2 = -3, \quad s_3 = -4, \quad s_4 = -5,$$

the coefficients are calculated as:

$$a_1 = 14, \quad (40)$$

$$a_2 = 49, \quad (41)$$

$$a_3 = -122, \quad (42)$$

$$a_4 = 120. \quad (43)$$

The gains are therefore:

$$k_4 = 14, \quad k_3 = 49, \quad k_2 = -122, \quad k_1 = 120.$$

IV. RESULTS

A. open-loop response

We first computed the system as a matlab function, and ran the simulation to give out the open loop response plot, showing the rod's angular position with time, as shown in figure 9, with $x_0 = [\pi/2 \ 0 \ 0 \ 0]$ and 0 input as initial system conditions.

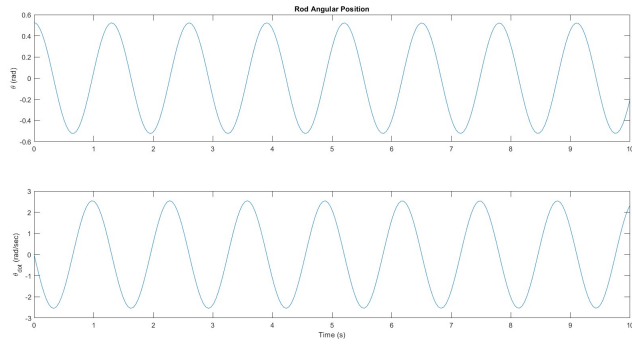


Fig. 8. The model's open-loop response plot

B. closed-loop response

The red curve begins at 1.6 and rapidly decreases to settle at 0 within approximately 2–3 seconds. This behavior represents an exponential decay, typical of a stable system. The output (θ , angular position) starts from a non-zero initial condition (1.6) and stabilizes to zero.

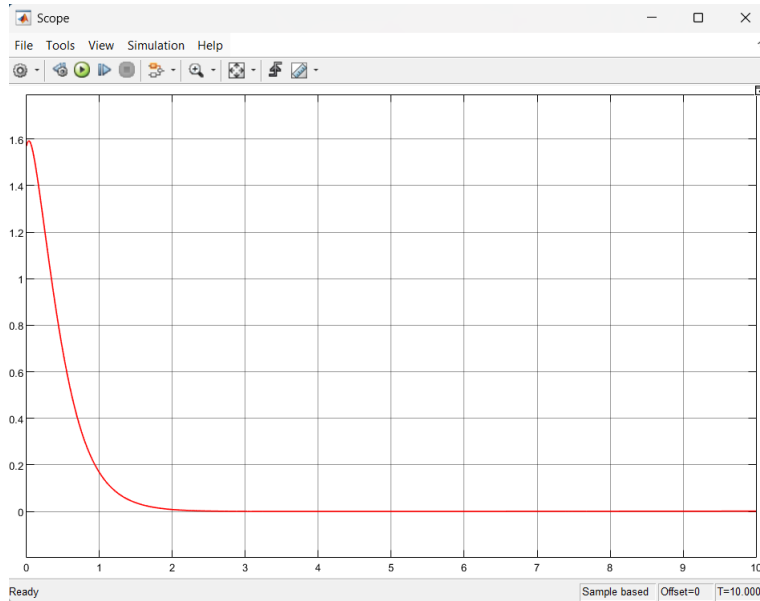


Fig. 9. The model's closed response plot

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